Answers and talk notes

Explore Numerical Expressions with Exponents

What your student is learning

Tonight we evaluate powers — writing them out as repeated multiplication and finding the value.

Answers

Ask your student to explain before you look.

1. **B.** 5. In 8^5 there are 5 factors of 8, because the exponent counts the copies of the base.
2. **A.** 32. $2 \times 2 \times 2 \times 2 \times 2 = 2^5 = 32$. Doubling 5 times: 2, 4, 8, 16, 32.
3. **C.** 125. $5^3 = 5 \times 5 \times 5 = 25 \times 5 = 125$.
4. **D.** 2^4. $2^4 = 16$, which is greater than $3^2 = 9$, $2^3 = 8$ and $1^5 = 1$.
5. **B.** 49. $7^2 = 7 \times 7 = 49$.
6. **A.** 10,000. $10^4 = 10 \times 10 \times 10 \times 10 = 10{,}000$ cells.

Talk about it

How many copies of the base did you multiply, and how do you know from the exponent?

Home mission

Use coins or buttons. Put out 3 for round 1. For round 2, add 3 new ones for each one. Count. Then say the powers aloud: 3 to the 1st, 2nd, 3rd. Predict round 4 before you build it.

When they are stuck, ask

- Can you write this out as repeated multiplication first?
- How many copies of the base are there?
- Are we multiplying BY the small number, or multiplying copies together?

Try not to say

- Multiply the base by the exponent.
- Squared and times two are the same thing.
- Do it all in one step in your head.